INCH-POUND

MIL-P-70819 (AR)
28 February 1991

MILITARY SPECIFICATION

PRIMER, ELECTRIC, M128
LOADING, ASSEMBLING AND PACKING

This Specification is approved for use within the U.S. Army Armament, Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification contains requirements and provides quality assurance provisions for loading, assembling and packing of the M128 Electric Primer.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications standards and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

MILITARY

MIL-P-116	Preservation, Methods of.
MIL-B-117	- Bags, Sleeves and Tubing - Interior Packging.
MIL-P-223	- Powder, Black.
MIL-A-48078	- Ammunition, Standard Quality Assurance Provisions, General Specification for.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, U.S. Army ARDEC, ATTN: SMCAR-BAC-S, Picatinny Arsenal, New Jersey 07806-5000 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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STANDARDS

MILITARY

MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-286	- Propellants, Solid: Sampling, Examination and Testing Inspection by Attributes.
MIL-STD-1234	- Pyrotechnics, Sampling Inspection and Testing.

(Unless otherwise indicated, copies of federal and military specifications, standards and handbooks are available from the Standardization Documents Order Desk, Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWINGS

ARMY

8847545	- Box, Fiberboard, Packing, Ammunition for Primer, Electric.
8847546	- Box, Packing, Ammunition, for Primer, Electric.
12910250	- Primer, Electric, Loading Assembly, M128.

(Copies of Government documents, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

PUBLICATIONS

(Title) 49 Code of Federal Regulations
Transportation of Hazardous Materials.

(Application for copies should be addressed to the Regulations Managenent Corp., 4 Old Red Lion Rd., Vincentown NJ, 08088-8632.)

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2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Material. Materials and parts shall be in accordance with applicable drawings and specifications.

3.2 Primers. The loaded primers shall comply with all requirements specified on drawing 12910250 and applicable specifications.

3.3 Moisture content.

3.3.1 Ignition element charge. The moisture content of the ignition element charge composition after drying shall not exceed 0.02 percent.

3.3.2 Black powder. The moisture content of the black powder at the time of loading the closing plug shall not exceed 0.50 percent.

3.3.3 Benite. The moisture content of the benite at the loading station at the time of loading shall not exceed 1.00 percent.

3.4 Ignition element charge composition. The chemical composition of the ignition charge shall be:

Component	Percentage
Potassium Chlorate and Gum Solids	46.1 ± 2.1%
Lead Thiocyanate	35.5 ± 2.0%
Charcoal	20.0 ± 1.5%

3.5 Lacquer in flash holes. The lacquer in primer body flash holes shall comply with illustration 1, 2, or 3 of Figure 1.

3.6 Assembly torque. Threaded joints shall withstand subsequent reapplication of the minimum assembly torque specified on the applicable drawing, without moving, while thread sealing compounds are wet.

3.7 Electrical resistance test. The electrical resistance of the head loading assembly when measured between the primer head (dwg. 9327339) and the electrode (dwg. 8839507) shall not be less than 0.80 ohms or more than 1.80 ohms.

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3.8 Functioning/non-functioning test of head loading assembly.

3.8.1 Functioning. The head loading assembly shall function when a current of 1.25 amperes, maximum, is applied.

3.8.2 Non-functioning. The primer loading assembly shall not function when a current of 190 to 200 milliamperes is applied.

3.9 Proving ground. The primer shall comply with the following requirements when fired ballistically:

a. The primer shall not fail to fire.
b. The primer body shall not rupture or become detached.
c. There shall be no audible hangfire(s).
d. The primer shall not squib.
e. The tip shall not become separated from the body.

3.10 Check test for possible deterioration of primers. Check tests are designed to prevent loading ammunition items with primers which have deteriorated. They should be performed on each lot of primers which have been stored for more than two years or which have been subjected to any adverse conditions, however brief, within that period of time. The following tests are specified:

a) Moisture Content of Benite
b) Electrical Resistance
c) Functioning
d) Non-Functioning

3.11 First article inspection. This specification contains technical provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

3.12 Workmanship. All parts and assemblies shall be fabricated and loaded in accordance with the applicable specifications and drawings. They shall be clean and free of burrs, sharp edges, unblended radii, surface defects, chips, dirt, grease and oil (except where specifically required), corrosion products and other foreign matter. The cleaning method used shall not be injurious to any part of the assembly nor shall they be contaminated by the cleaning agent. Exterior surface coatings shall be continuous except for a few light scratches not exposing base material. All required markings shall be neat and sharply defined. Required packing shall be dry.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.2 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in this specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.3 Inspection equipment. Inspection equipment required to perform the examinations and tests prescribed herein is described in the "Inspection Method Reference" column in the tables starting with 4.4.2

4.4 Classification of inspections. The following types of inspections shall be conducted on this item:

a. First article inspection
b. Quality Conformance Inspection

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4.7 Methods of Inspection

4.5 First article inspection

4.5.1 Submission. The contractor shall submit a first article sample as designated by the Contracting Officer for evaluation in accordance with provisions of 4.5.2. The first article sample shall consist of the following items in sample quantities indicated.

Part/Assembly Description	Drawings	Quantity
Element, Ignition, Electric Loading Assembly	8839505	10
Closing Plug Loading Assembly	9327346	10
Head Loading Assembly	9327345	400
Primer, Electric, M128	12910250	100
Box, Fiberboard, Packing, Ammunition for Primer, Electric	8847545	5
Box, Packing, Ammunition for Primer, Electric	8847546	5

4.5.2 Inspections to be performed. See MIL-A-48078 and Table I specified herein.

4.5.3 Rejection. See MIL-A-48078.

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Table I. First article inspection.

CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE Primer Assembly & Components	SHEET 1 OF 2	DRAWING NUMBER See Below
			NEXT HIGHER ASSEMBLY

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
	Element, Ignition, Electric Loading Assembly (Dwg. 8839505)			
	Examination for defects	10	3.1	4.6.2.3
	Closing Plug Loading Assembly (Dwg.9327346)			
	Examination for defects	10	3.1	4.6.2.4
	Head Loading Assembly (Dwg. 9327345)			
	Examination for defects	400	3.1	4.6.2.5
	Functioning	200*	3.8.1	4.8.5
	Non-functioning	200*	3.8.2	4.8.6
	Primer, Electric, M128 Loading Assembly (Dwg. 12910250)			
	Examination for defects	50**	3.2	4.6.2.6, 4.6.2.7, 4.6.2.8
	Proving Ground	50**	3.2	4.8.8

NOTES: * Should any sample fail to meet the applicable requirements, the first article sample shall be rejected.
** If a critical defect occurs or if 2 or more major defects occur, the first article sample shall be rejected.

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Table I. First article inspection.

CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE: Primer Assembly & Components	SHEET 2 OF 2	DRAWING NUMBER: See Below
			NEXT HIGHER ASSEMBLY

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
	Barrier Bag (Sealed) (Dwg. 8847545, MIL-B-117) Examination for defects	5	3.1	4.6.2.9
	Box, Fiberboard, Packing, Ammunition for Primer, Electric (Dwg. 8847545) Examination for defects	5	3.1	4.6.2.10, 4.6.2.11
	Box, Packing, Ammunition for Primer, Electric (Dwg. 8847546) Examination for defects	5	3.1	4.6.2.12, 4.6.2.13

NOTES:

AMSMC Form 1570b, 1 Jul 89

Replaces 1570, 1 Feb 85, which may not be used.

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4.6 Quality conformance inspection

4.6.1 Inspection lot formation. Inspection lots shall comply with lot formation provisions of MIL-A-48078. In addition, inspection lots of loaded primers shall contain the following components from one lot interfix from one manufacturer: ignition element, metal parts, charge composition, and benite. Black powder shall be from one lot only.

4.6.2 Examination and tests.

4.6.2.1 Classification of characteristics. Quality conformance examinations and tests are specified in the following Classification of Characteristics paragraphs. The contractor's quality program or detailed inspection plan shall provide assurance of compliance of all characteristics with the applicable drawing and specification requirements utilizing as a minimum the conformance criteria specified herein. Unless otherwise specified herein, attributes sampling inspection shall be conducted in accordance with Table II below using Level I for major defects and Level II for minor defects. Samples shall be selected at random.

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Table II Sampling
Sampling Plan for Quality Conformance
(Accept on Zero Defects)

Lot Size	Sample Size (Characteristic)	
	Major	Minor
2 to 8	*	*
9 to 150	32	13
151 to 500	32	30
501 to 1200	74	35
1201 to 3200	74	43
3201 to 10000	87	50
10001 to 35000	109	61

* 100% inspection
NOTE: If sample size equals or exceeds lot or batch size, do 100% inspection.

4.6.2.2 Alternative inspection provisions. Alternative inspection procedures, methods or equipment, such as statistical process control, tool control, other types of sampling procedures, etc., may be used by the contractor when they provide, as a minimum, the level of quality assurance required by the provisions specified herein. Prior to applying such alternative procedures, methods, or equipment, the contractor shall describe them in a written proposal submitted to the procuring contracting officer for evaluation and approval by the Government (see 6.4). When required, the contractor shall demonstrate that the effectiveness of the proposed alternative(s) is equal to or better than the specified quality assurance provisions herein. In cases of dispute as to whether the contractor's proposed alternative(s) provide equal assurance, the provisions of this specification shall apply. All approved alternative inspection provisions shall be specifically incorporated into the contractor's quality program or detailed inspection system, as applicable.

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QUALITY CONFORMA' ` INSPECTION

CLASSIFICATION OF CHARACTERISTICS

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PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.6.2.3	Element, ignition, electric, loading assembly	1 OF 1	8839505
			NEXT HIGHER ASSEMBLY 12910250

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical				
1	Charge above flush	100%	3.2	Visual
2	Charge composition on outside surface	100%	3.2	Visual
Major				
101	Charge missing or insufficient	Table II	3.2	Visual
102	Moisture content	4.7.1.1	3.3.1	4.8.1.1
103	Analysis of ignition element charge composition	4.7.2	3.4	4.8.2
Minor				
201	Workmanship	Table II	3.12	Visual

NOTES:

AMSMC Form 1570b, 1 Jul 89 Replaces 1570, 1 Feb 85, which may not be used.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF CHARACTERISTICS

MIL-P-70819 (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.6.2.4	Closing plug loading assembly	1 OF 1	9327346
			NEXT HIGHER ASSEMBLY 12910250

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Black powder charge missing	100%	3.2	Visual
102	Disc(s) missing , punctured or peeling, (when applicable)	Table II	3.2	Visual
103	Lacquer missing from disc (when applicable)	Table II	3.2	Visual
104	Tape missing or improperly assembled (when applicable)	Table II	3.2	Visual
105	Moisture content of black powder charge	4.7.1.2	3.3.2	4.8.1.2
Minor				
201	Workmanship	Table II	3.12	Visual

NOTES:

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QUALITY CONFORMAN INSPECTION

CLASSIFICATION OF CHARACTERISTICS

MIL-P-70819 (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.6.2.5	Head loading assembly	1 OF 1	9327345
			NEXT HIGHER ASSEMBLY 12910250

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical				
1	Non-functioning test	4.7.4	3.8.2	4.8.6
Major				
101	Depth to electrode	Table II	3.2	Gage
102	Electrode resistance test	100%	3.7	4.8.3
103	Flashing hole obstructed	Table II	3.2	Gage
104	Functioning test	4.7.3	3.8.1	4.8.5
Minor				
201	Lacquer missing from closing plug threads	Table II	3.2	Visual
202	Workmanship	Table II	3.12	Visual

NOTES:

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CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.6.2.6	Primer assembly, prior to loading	1 OF 1	12910250
			NEXT HIGHER ASSEMBLY

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major 101	Component damaged	Table II	3.2	Visual
Minor 201	Workmanship	Table II	3.12	Visual

NOTES:

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CLASSIFICATION OF CHARACTERISTICS

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PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.6.2.7	Primer assembly prior to assembly of tip	1 OF 1	12910250
			NEXT HIGHER ASSEMBLY

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Lacquer missing from tip threads	Table II	3.2	Visual
102	Charge missing	Table II	3.2	Visual
103	Liner improperly positioned	Table II	3.2	Visual
Minor				
201	Workmanship	Table II	3.12	Visual

NOTES:

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CLASSIFICATION OF CHARACTERISTICS

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PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.6.2.8	Primer, electric, M128	1 OF 1	12910250
			NEXT HIGHER ASSEMBLY

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical				
1	Body ruptures or becomes detached	4.7.5	3.9	4.8.8
2	Tip separates from body	4.7.5	3.9	4.8.8
3	Tip missing	100%	3.2	Gage
Major				
101	Liner improperly positioned (prior to loading)	Table II	3.2	Visual
102	Flash hole sealing improper (prior to loading)	Table II	3.2	Visual
103	Weight of benite charge	Table II	3.2	Visual
104	Length of benite charge	Table II	3.2	Scale
105	Lacquer missing from tip threads	Table II	3.2	Gage
106	Sealant missing from body threads	Table II	3.2	Visual
107	Body fails to withstand reapplication of assembly torque	Table II	3.6	4.8.4
108	Moisture content of benite	4.7.1.3	3.3.3	4.8.1.3
109	Proving ground test - failure to fire	4.7.5	3.9	4.8.8
110	Proving ground test - assembly hangfires	4.7.5	3.9	4.8.8
111	Proving ground test - assembly squibs	4.7.5	3.9	4.8.8
112	Electrical resistance test	100%	3.7	4.8.8
113	Length, total, max.	100%	3.2	Gage
Minor				
201	Depth to tip	Table II	3.2	Gage
202	Marking incorrect or unidentifiable	Table II	3.2	Visual
203	Workmanship	Table II	3.12	Visual

NOTES:

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MIL-P-70819 (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.6.2.9	Barrier bag (sealed)	1 OF 1	8847545 (MIL-B-117)
			NEXT HIGHER ASSEMBLY

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Bag, punctured, torn or cut	Table II	3.2	Visual
102	Seal mislocated or incomplete	Table II	3.2	Visual
103	Heat seal test of barrier bag	4.7.6	3.2	Gage
Minor				
201	Workmanship		3.12	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.6.2.10	Box, fiberboard (before sealing)	1 OF 1	8847545
			NEXT HIGHER ASSEMBLY

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Any packing material missing	Table II	3.2	Visual
102	Separator(s) damaged to the extent that primers touch each other	Table II	3.2	Visual
Minor				
201	Workmanship	Table II	3.12	Visual

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CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.6.2.11	Box, fiberboard (sealed)	1 OF 1	8847545
			NEXT HIGHER ASSEMBLY

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major	None defined			
Minor				
201	Sealing strip torn, badly wrinkled, incomplete	Table II	3.2	Visual
202	Box torn, cut or punctured	Table II	3.2	Visual
203	Contents loose to an extent greater than permitted by packing design	Table II	3.2	Manual
204	Marking missing, misleading or unidentifiable	Table II	3.2	Visual
205	Workmanship	Table II	3.12	Visual

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF CHARACTERISTICS

MIL-P-70819 (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.6.2.12	Box (before sealing)	1 OF 1	8847546
			NEXT HIGHER ASSEMBLY

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major 101	Box, fiberboard, missing	Table II	3.2	Visual
Minor 201	Workmanship	Table II	3.12	Visual

NOTES:

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CLASSIFICATION OF CHARACTERISTICS

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PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.6.2.13	Box (sealed)	1 OF 1	8847546
			NEXT HIGHER ASSEMBLY

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Protruding nail	Table II	3.2	Visual
102	Board broken or loose	Table II	3.2	Visual/Manual
103	Strapping missing, broken or loose	Table II	3.2	Visual/Manual
Minor				
201	Contents loose	Table II	3.2	Manual
202	Strapping improperly engaged or mislocated	Table II	3.2	Visual
203	Marking missing, misleading or unidentifiable	Table II	3.2	Visual
204	Workmanship	Table II	3.12	Visual

NOTES:

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4.7 Testing

4.7.1 Moisture content.

4.7.1.1 Ignition element charge. Three samples minimum shall be selected from each drying oven after completion of the drying cycle. If the moisture content of any sample exceeds the requirement, that quantity of material represented by the sample shall either be redried until acceptable results are achieved, or rejected. The selected samples must not be dried more that any other unit which could have been selected.

4.7.1.2 Black powder. One sample, minimum, shall be selected from each eight hours production. If the moisture content of the sample exceeds the requirement, and loading has not begun, that quantity of material represented by the sample shall be rejected. If assemblies have been loaded with material containing excessive moisture, the remaining material represented by the sample, together with all assemblies loaded with the non-conforming material shall be rejected.

4.7.1.3 Benite. One 100 gram sample of benite shall be selected at the loading station, during each eight hours' production. If the moisture content of the sample exceeds the requirement, the primers loaded since the last acceptable sample shall be rejected.

4.7.2 Analysis of ignition element charge. Approximately two grams of ignition charge shall be selected from that quantity of charge used in loading a complete lot of igniters. The lot shall be rejected if it fails to comply with the requirements.

4.7.3 Functioning. One hundred and twenty five samples shall be selected from each lot for this test. If one or more failures occur, the lot shall be rejected. The functioning test is considered to be a destructive test. Assemblies subjected to this test shall not be returned to the lot.

4.7.4 Non-functioning. One hundred and twenty five samples shall be selected from each lot for this test. If any assembly functions, the lot shall be rejected. The non-functioning test is considered to be a destructive test. Assemblies subjected to this test shall not be returned to the lot.

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4.7.5 Proving ground.

4.7.5.1 Initial lots. Beginning with the first lot produced and continuing until three consecutive lots have complied with the acceptance criteria specified, thirty-five primers shall be randomly selected from each lot for this test. If a critical defect occurs or one or more major defects occur, the lot shall be rejected.

4.7.5.2 Subsequent lots. After three consecutive lots have met the criteria specified in 4.7.5.1, ten primers shall be randomly selected from each lot for test. If one critical defect occurs or if one or more major defects occur, the lot shall be rejected.

4.7.6 Check test

4.7.6.1 Moisture content of benite. A sample of 25 primers shall be selected for this test. If one or more primers fail to comply with requirements, the lot shall be rejected.

4.7.6.2 Electrical resistance. This test shall be performed 100 percent. Any primer assembly which fails to comply with the requirement shall be classed defective and removed from the lot.

4.7.6.3 Functioning. 300 primers shall be randomly selected from each lot for this test. If three or more assemblies fail to comply with the requirement the lot shall be rejected.

4.7.6.4 Non-functioning. 50 primers shall be selected from each lot for this test. If one or more assemblies fail to comply with the requirement the lot shall be rejected.

4.7.7 Heat seal test of barrier bag. Sampling and acceptance criteria shall be as prescribed in MIL-P-116 for the "Heat Seal Test".

4.8 Methods of inspection.

4.8.1 Moisture content

4.8.1.1 Ignition element charge composition. Moisture content of the charge composition shall be determined as specified in Method 102.2.1 of MIL-STD-1234 except that only a two hour dry time shall be used. The upper insulator may be removed from the assembly prior to analysis.

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4.8.1.2 Black powder. Moisture content of the black powder shall be determined in accordance with the procedure specified in MIL-P-223.

4.8.1.3 Benite. Moisture content of the benite shall be determined in accordance with Method 102.1 of MIL-STD-286.

4.8.2 Analysis of ignition charge composition.

4.8.2.1 Charcoal. Transfer approximately 2 grams of the igniter charge composition into a tared medium porosity crucible. Place in oven 70° C overnight to drive off water. Place crucible in desiccator to cool and then weigh to establish sample weight. Wash with ten 25 ml portions of a 5 percent potassium nitrate solution followed by 50 ml of distilled water. Use a small glass rod to break up any lumps in the sample during the extraction process. Preserve the filtrate for the determination for lead thiocyanate. Place in oven maintained at 100 ± 5° C for 2 hours. Cool in a desiccator and weigh. Calculate the percent charcoal as follows:

$$\text{Percent charcoal} = \frac{(100)\ (C - B)}{A - B}$$

A = Sample weight before washing (including crucible)
B = Empty crucible weight
C = Sample weight after washing

4.8.2.2 Lead thiocyanate. Exactly 35 ml of standardized 0.1 normal (N) silver nitrate shall be added to the filtrate reserved from the charcoal determination. The flask and contents shall be swirled and 100 ml of 3N nitric acid shall be added. Nitric acid used should be water white and free of oxides of nitrogen. The swirling of the mixture shall be continued for five minutes, 5 ml of a saturated solution of ferric ammonium sulfate shall be added and the solution titrated with standardized 0.1N potassium thiocyanate solution to the first permanent pale rose coloration. The percentage of lead thiocyanate in the sample shall be calculated on a dry basis as follows:

$$\text{Percent lead thiocyanate} = \frac{1.617\ (A - B)}{C}$$

A = Volume of silver nitrate solution
B = Volume of potassium thiocyanate solution
C = Weight of sample

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4.8.2.3 Potassium chlorate and gum solids. The percentage of potassium chlorate and gum solids is obtained by subtracting the total percentage of charcoal and lead thiocyanate from one hundred percent.

4.8.3 Electrical resistance. The head loading assembly (dwg. 9327345), shall be tested for electrical resistance by applying an electrical current of 20 milliamperes, maximum, using the electrode as one contact point and the head as the other.

4.8.4 Torque testing The torque test shall be performed while the lacquer is still wet (within 5 minutes after the lacquer application). The head shall be held in a suitable fixture, and the torque applied in the same direction as required for initial tightening. The joint under test shall be marked whenever is possible to provide a means of detecting any movement between adjacent parts. Any discernible movement between adjacent parts shall be regarded as a torque test failure.

4.8.5 Functioning. The primer loading assembly shall be tested for electrical functioning by applying an electrical current of 1.25 amperes, maximum, using the electrode for one contact point and the head for the other.

4.8.6 Non-functioning. The head loading assembly shall be tested for non-functioning by applying an electrical current of 190 to 200 milliamperes for one second minimum, using the electrode as one contact point and the head as the other.

4.8.7 Heat seal test of barrier bag. The test procedure shall be as prescribed in MIL-P-116 for the "Heat Seal Test".

4.8.8 Proving ground. The primers shall be assembled into complete rounds utilizing M900 or equivalent slug projectiles for which the primer is standard and fired from an M68A1 gun. The test shall be fired with a standard service charge at $120^{\circ} - 5^{\circ}$F at resultant pressures.

The primer shall be observed for evidence of the following defects:

TABLE III

Defects	Classification
Body ruptures or becomes detached	Critical
Tip separates from body	Critical
Failure to fire	Major
Assembly hangfires	Major
Assembly squibs	Major

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5. PACKAGING

5.1 Preservation and Packaging. Primers shall be packed in accordance with 8847545 and (Title) 49CFR Parts 100-199, Para. 173.107. In addition the fiberboard box must have appropriate separators and cushions to provide a tight pack and provide satisfactory physical protection to the primer. The fiberboard box must be enclosed in a barrier bag conforming to MIL-B-117, Type 1, Class E, to provide satisfactory moisture protection to the primer. The barrier bag shall be heat-sealed in accordance with MIL-B-117.

5.2 Packing. All packing must meet requirements of 8847546 and (Title) 49CFR parts 100-199, Para. 173.107. Appropriate filler material shall be added to sides, ends, top and bottom of interior of pack to provide a tight pack.

5.3 Marking. Marking must be in accordance with MIL-STD-129 and (Title) 49CFR Parts 100-199 Para. 173.107.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The M128 Primer is intended for use in the M900 105MM APFSDS-T Cartridge.

6.2 Acquisition requirements. Procurement documents must specify the following:

a. Title, number and date of this specification.

b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1).

6.3 Precaution. This specification covers sampling and testing of chemical, toxic or explosive materials which are potentially hazardous to personnel. Accordingly it is emphasized that all applicable safety rules, regulations and procedures must be followed in handling and processing these materials.

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6.4 Submission of inspection equipment designs for approval. The contractor shall submit for approval, inspection equipment designs in accordance with the terms of the contract. See MIL-A-48078, Section 6. Submit designs to:

Commander
U.S. Army Armament Research, Development
and Engineering Center
ATTN: AMSMC-QAT-I (D)
Picatinny Arsenal, NJ 07806-5000

This address will be specified on the Contract Data Requirements List in the contract.

6.5 Submission of firing records and ammunition data cards. In addition to the normal distribution of records, when the M128 Primer is procured by the Army Armament, Munitions and Chemical Command, one copy of all firing records and ammunition data cards shall be forwarded to:

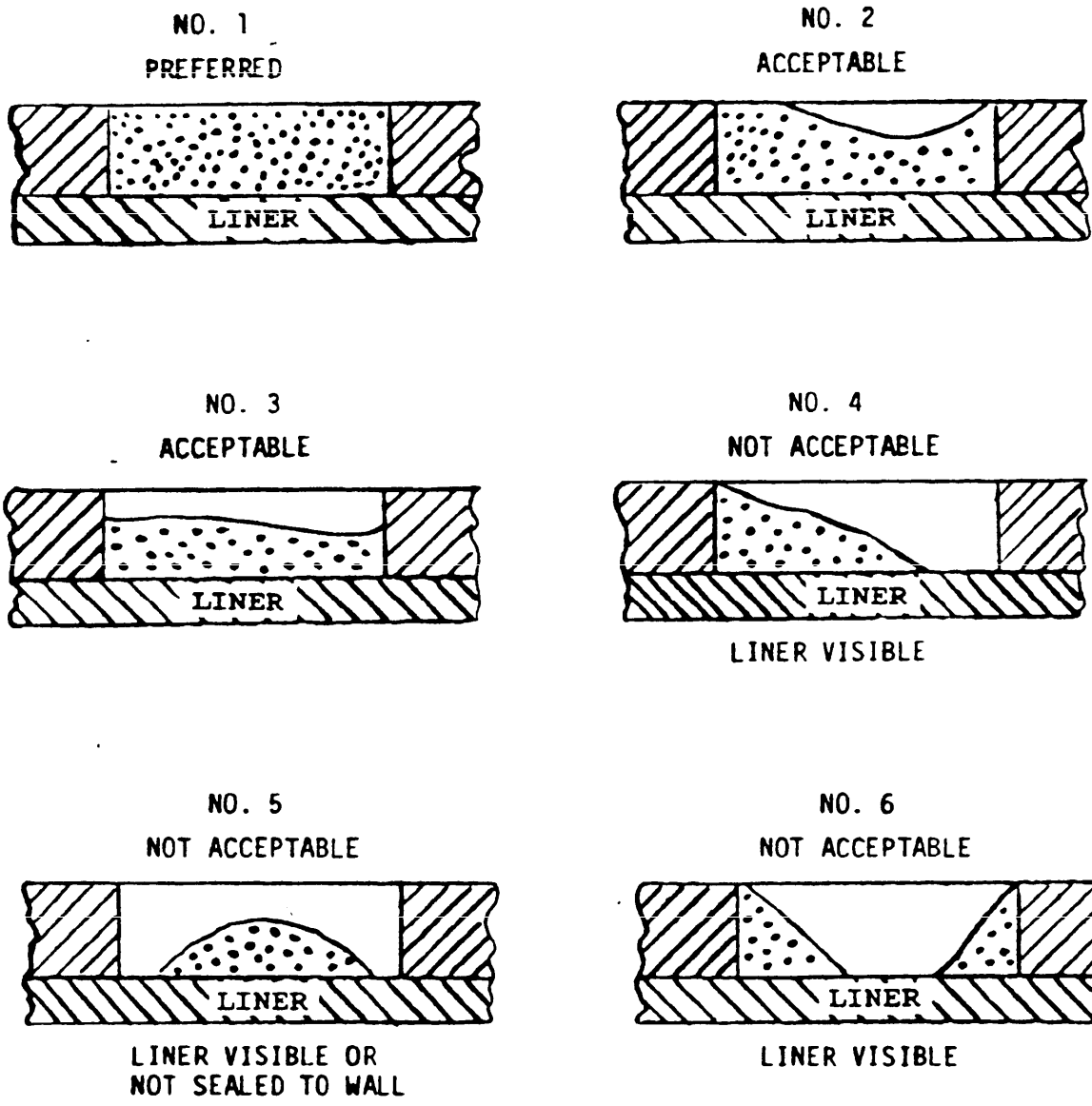
Commander
U.S. Army Armament Research, Development
and Engineering Center
ATTN: AMSMC-QAT-A (D)/SMCAR-CCH-P
Picatinny Arsenal, NJ 07806-5000

6.6 Submission of alternative inspection provisions. Proposed alternative inspection provisions should be submitted by the contractor to the procuring contracting officer for evaluation and approval by the technical activity responsible for preparation of this specification.

6.7 Subject term (key Word) listing.

Ammunition
Tank
Cannon
Cartridge
Primer

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Standards for Lacquer Conditions in Flash Holes

Figure 1

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Custodian:
Army - AR

Preparing activity:
Army - AR

(Project 1390-A678)

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